Commission believes that the expanded definition of core data will be useful to market participants and will help fulfill needs that are not currently being met by SIP data. Additionally, and for the same reasons, the Commission disagrees with comments suggesting that proposed core data would not be useful to many market participants, that proprietary market data products are adequately meeting the needs of all market participants, and that all market participants that have a need to access these products are able to do so. Rather, the definition of core data specifies important information that would be useful to a wide variety of market participants—including those who do not obtain it through proprietary market data products today—and facilitates a broader dissemination of this information.[167]

In addition, the Commission disagrees with comments that the definition of core data would require market participants, including non-professional investors, to purchase or consume all data that would be defined as core data, and thereby increase the cost of core data for all.[168] Competing consolidators are not required to offer a data product that includes all consolidated

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stocks, depth of book data, and auction information to help market participants use core data to trade in a more informed and effective manner in light of these developments); infra Sections II.D through II.G.

[167] See supra note 114 (describing the authority under Section 11A of the Exchange Act to specify additional information that must be made available within the national market system); Section I.A (explaining the need to improve and modernize the national market system to fulfill the goals of Section 11A of the Exchange Act and to meet the current core data needs of all market participants). As stated below, some market participants stated that those who do not buy the exchange proprietary DOB feeds and associated connectivity and transmission offerings are at a competitive disadvantage relative to market participants who purchase these feeds. See infra note 1620 and accompanying text. See also infra Sections III.E.2(c); V.C.2(b)(i)a (discussing how the amendments will affect data content fees).

[168] See TD Ameritrade Letter at 3; NYSE Letter II at 3–4; Nasdaq Letter IV at 7–8.

market data,[169] and the Commission has explicitly stated that the proposed definitions of core data and consolidated market data do not "mandat[e] the consumption" of particular data elements.[170] Thus, the Commission believes it has considered and addressed the needs of market participants that do not directly need all elements of core data. The purpose of expanding core data is to promote wider dissemination of data that will be useful in meeting the needs of a broad array of market participants. As explained below, the enhanced core data content will benefit all investors, regardless of whether they directly consume it.[171] Furthermore, the Operating Committee of the effective national market system plan(s) could develop fees for data content underlying consolidated market data offerings for different subsets of consolidated market data to suit the needs of various market participants, as one member of the Operating Committee has already suggested.[172] Within this framework, the Commission believes that the market would develop to enable market participants to consume and pay for the market data that best suits their needs and that there would be downward pressure on data content fees.[173]

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169 See infra Section III.C.8(a).

170 Proposing Release, 85 FR at 16775.

171 See infra notes 174–176 and accompanying text.

172 See letter from Elizabeth K. King, Chief Regulatory Officer, ICE, General Counsel and Corporate Secretary, NYSE to Vanessa Countryman, Secretary, Commission, dated Feb. 5, 2020, ("Feb. NYSE Letter") (recommending that the Commission expand SIP content and "create products designed for modern use cases, including a SIP product with depth-of-book quotes for institutional traders and a National Best Bid and Offer ('NBBO') only version for retail customers, with fees based on content entitlements (or levels) instead of user type"). See also infra notes 1201–1208 and accompanying text.

173 See supra note 28 (describing comments received by the Commission regarding the high cost of proprietary data products that contain data needed for effective participation in the markets); infra Sections III.E.2(c); V.C.2(b)(i)a (discussing how the amendments will affect data content fees).

Moreover, the Commission believes that all investors will benefit, directly or indirectly, from the expanded definition of core data. Even if only a subset of market participants may choose to acquire directly a data product that includes the full set of data elements included within the definition of core data, the Commission believes that there will be ample demand for expanded core data[174] and a corresponding incentive for competing consolidators to offer more content-rich products. The Commission expects that direct purchasers of such products likely will include many broker-dealers that are electronically routing orders for execution or executing orders internally. As discussed below, the additional data elements included within the definition of core data are useful to efficiently and effectively route and execute orders in today's dispersed electronic markets,[175] and their widespread availability should facilitate broker-dealers' ability to achieve best execution for customers.[176] Thus, broker-dealers will be incentivized to acquire products containing the expanded core data elements to compete effectively for customer business. In addition, by including these additional, important market data elements as part of expanded core data, this rulemaking should help facilitate executing broker-dealers' access to information, to the benefit of all investors. Accordingly, while the Commission expects only some market participants to choose to purchase a data product that includes the full set of core data, any market participant that submits an order in an NMS stock should benefit indirectly from their doing so because more executing broker-dealers will receive the data elements that will help them place customer orders in a more informed and effective manner.

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174 See infra notes 878–880; supra notes 163–167 and accompanying text.

175 See infra notes 878–880 and accompanying text.

176 See supra Section I.E.

Finally, in response to the comment recommending that a "retail interest indicator" be added to quotes,[177] the definition of self-regulatory organization-specific program information already incorporates retail interest indicators disseminated in current SIP data and established pursuant to exchange retail liquidity programs in the definition of consolidated market data.[178]

(b) Odd-Lot Quotations

In the Proposing Release, the Commission solicited comment on whether core data should include odd-lot quotations, but did not include odd-lot quotes in the definition of core data other than by incorporating them through the proposed definition of round lot.[179] Several commenters recommended directly including odd-lots in core data rather than doing so through the mechanism of the proposed definition of round lot.[180] Specifically, one commenter suggested including odd-lots priced better than the PBBO in core data,[181] and another suggested including the best-priced odd-lot quotation from each exchange.[182] Another commenter supported the Commission's aim of increasing odd-lot transparency for higher priced securities

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177 See Clearpool Letter at 6.

178 See infra Section II.K.

179 See Proposing Release, 85 FR at 16746.

180 See letter from Patrick Sexton, Executive Vice President, General Counsel, and Corporate Secretary, Cboe, to Vanessa Countryman, Secretary, Commission, dated May 26, 2020, ("Cboe Letter") at 15; NYSE Letter II at 5; Nasdaq Letter IV at 14; RBC Letter at 5; letters to Vanessa Countryman, Secretary, Commission, from Kimberly Unger, Chief Executive Officer and Executive Director, STANY, dated June 11, 2020, ("STANY Letter II") at 3; Anders Franzon, General Counsel, MEMX LLC, dated May 26, 2020, ("MEMX Letter") at 2 ("[A]ll data currently made available through proprietary data feeds should be available through NMS data feeds. This includes complete depth-of-book data (and thus all odd lot data)….").

181 See CBOE Letter at 15.

182 See NYSE Letter II at 5.

but questioned doing so through the proposed definition of round lot.[183] One commenter recommended adding unprotected odd-lots to core data, combined with best execution guidance on broker-dealer obligations with respect to odd-lot quotations, rather than redefining round lot.[184] Similarly, another commenter recommended including odd-lot quotations in core data while leaving the definition of round lot as it currently stands.[185] A different commenter recommended delaying odd-lots to mitigate the impact on processing times.[186] On the other hand, one commenter expressed concerns that adding odd-lot quotations to core data would harm investor confidence in the markets resulting from confusion over protected and unprotected quotes and increased costs and latency for core data by adding more information that needs to be disseminated.[187] A different commenter presented data showing that, for a significant percent of orders in each of the Commission's proposed round lot tiers, there would still be a contra-side odd-lot quote better than the NBBO.[188]

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183 See Nasdaq Letter IV at 14.

184 See RBC Letter at 5.

185 See STANY Letter II at 3.

186 See Data Boiler Letter I at 19.

187 See TD Ameritrade Letter at 4–5.

188 Memorandum from the Division of Trading and Markets regarding a June 19, 2020, meeting with representatives of JP Morgan ("JP Morgan Memo to File") at 2 ("Under today's rules: 11.6% of orders contain a contra-side oddlot [sic] quote better than the NBBO. Under SEC's proposed round lot parameters: (i) Bucket A ($50.00 and less) – 100 share round lot – 6.3% of orders would still contain a contra-side oddlot [sic] quote better than the NBBO; (ii) Bucket B (between $50.01 and $100.00) – 20 share round lot – 10.9% of orders would still contain a contra-side oddlot [sic] quote better than the NBBO; (iii) Bucket C (between $100.01 and $500.00) – 10 share round lot – 11.6% of orders would still contain a contra-side oddlot [sic] quote better than the NBBO; (iv) Bucket D (between $500.01 and $1,000.00) – 2 share round lot – 23% of orders would

The Commission continues to be concerned that the availability of odd-lot order information solely to market participants who have purchased proprietary market data products creates a potentially significant information asymmetry relative to market participants who purchase only SIP data.[189] For the reasons discussed below, the Commission is also modifying the definition of round lot.[190] While the proposed definition of round lot, as modified, would incorporate a substantial proportion of odd-lot quotations that occur at a price better than the NBBO for certain higher-priced stocks, the Commission is concerned that a significant amount of liquidity that could be available at better prices would be excluded from core data.[191] After considering comments, and given that the adopted round lot definition, on its own, would have resulted in less odd-lot information being included in core data, the Commission is adopting a definition of core data that includes all odd-lots that are priced at or better than the NBBO, aggregated at each price level at each national securities exchange and national securities association.

As summarized in Tables 1 and 2 below, staff analyzed data on the portion of all corporate stock and ETF volume executed on an exchange, transacted in a quantity less than 100 shares, at a price better than the prevailing NBBO, occurring in a quantity that would be defined as a round lot under both the adopted and proposed definitions of round lot.

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still contain a contra-side oddlot [sic] quote better than the NBBO; (v) Bucket E ($1,000.01 and higher) – 1 share round lot – all quotes are at round lot levels.”).

189 Proposing Release, 85 FR at 16741.

190 See infra Section II.D (explaining that the Commission is adopting a four-tiered definition of round lot rather than the five-tiered definition that was proposed).

191 See JP Morgan Memo to File at 2.

Table 1

Adopted Round Lot Tier	Adopted Round Lot Definition	Portion of all corporate stock and ETF volume executed on an exchange, transacted in a quantity less than 100 shares, at a price better than the prevailing NBBO, occurring in a quantity that would be defined as a round lot under the adopted definition of round lot
$0-$250.00	100 Shares	0 %
$250.01-$1,000	40 Shares	65.35%
$1,000.01-$10,000.00	10 Shares	88.28%
$10,000.01 or more	1 share	100.00%

Source: Equity consolidated data feeds (CTS and UTDF), as collected by MIDAS (May 2020); NYSE Daily TAQ.

Table 2

Proposed Round Lot Tier	Proposed Round Lot Definition	Portion of all corporate stock and ETF volume executed on an exchange, transacted in a quantity less than 100 shares, at a price better than the prevailing NBBO, occurring in a quantity that would be defined as a round lot under the proposed definition of round lot
$0-$50	100 shares	0%
$50.01-$100	20 shares	86.32%
$100.01-$500	10 shares	93.57%
$500.01-$1,000	2 shares	98.85%
$1,000.01 or more	1 share	100%

Source: Equity consolidated data feeds (CTS and UTDF), as collected by MIDAS (May 2020); NYSE Daily TAQ.

In comparison to the proposed tiers, the round lot tiers in the final rule would have excluded a significant proportion of better-priced odd-lot liquidity, particularly for stocks priced

between $50.01 and $250.00, and thus would not have included this liquidity in core data absent the Commission also including certain odd-lots in the definition of core data.

The Commission believes that this better-priced odd-lot liquidity needs to be reflected in core data because it will help investors and other market participants to trade in a more informed and effective manner and to achieve better executions and reduce the information asymmetries that currently exist between subscribers to SIP data and subscribers to proprietary data. However, the Commission continues to be concerned that adding all odd-lot quotations, particularly those at less aggressive price levels, could "burden systems, increase complexity, and degrade the usefulness of information in a manner that may not be warranted by the relative benefit of the additional information to investors and market participants" and that the inclusion of odd-lot quotations in proposed core data should be "reasonably calibrated."[192]

Therefore, the Commission is modifying the proposed definition of core data to include odd-lots that are priced at or more aggressively than the NBBO.[193] Specifically, pursuant to the

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192 Proposing Release, 85 FR at 16741.

193 As discussed below, the Commission is adopting a standard odd-lot aggregation methodology for all elements of core data, including the NBBO, wherein odd-lots across multiple price levels would be aggregated and disseminated at the least aggressive price. See infra Section II.C.2(d). As a result, odd-lots priced at or better than the NBBO could be both included in the NBBO and displayed in the aggregate at each price level by exchange. The Commission believes that this is appropriate, since the NBBO and odd-lot interest at or better than the NBBO provide independently valuable information to market participants. For example, odd-lots priced at or better than the NBBO are beneficial for order routing and achieving best execution, while the NBBO is protected under Rule 611 and must be provided in certain contexts pursuant to the Vendor Display Rule (Rule 603(c)). Additionally, as discussed below, competing consolidators will have the ability to customize data products for their customers, allowing investors to receive only the information they are able to process, so the Commission does not believe that including better-priced odd-lots both at each price level at each exchange and as part of an aggregated round lot would confuse investors.

revised definition of core data that the Commission is adopting, core data will include odd-lot quotations priced greater than or equal to the national best bid and less than or equal to the national best offer, aggregated at each price level at each national securities exchange and national securities association, in addition to odd-lot transaction data.[194] Making the best priced quotations available in core data is consistent with the Commission's goals in expanding the content of NMS information: enhancing the availability and usefulness of the information, reducing information asymmetries, and facilitating best execution. In addition, this modification is reasonably calibrated to include the odd-lot quotation data that would be of the most interest to investors and other market participants—namely, quotations that offer pricing at or superior to the NBBO—thus limiting complexity and systems burdens, and therefore costs, relative to alternatives such as including all odd-lot quotations.[195]

The Commission is also adopting the proposed inclusion of odd-lot transaction data in the definition of core data, through the definition of odd-lot information.[196] Odd-lot transaction data

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194 The Commission is adding odd-lots priced at or better than the NBBO through a new definition, "odd-lot information," that is included in the definition of core data. The definition of odd-lot information will include both odd-lots priced at or better than the NBBO and odd-lot transaction data. Odd-lot transaction data, which was added to SIP data by the national market system plans in 2013 (see Proposing Release, 85 FR at 16739), was proposed to be included in core data as a separate element, but the Commission believes it will simplify the definition of core data to include in a single defined term as "odd-lot information" odd-lots priced at or better than the NBBO and odd-lot transaction data. See infra Section VIII.

195 As discussed below, odd-lots priced less aggressively than the NBBO are not included in core data unless they aggregate to a round lot and are within the first five price levels after the NBBO. See infra Section II.F.2(e) (discussing odd-lot aggregation in the depth of book context).

196 See supra note 194.

is included in SIP data today, and it constitutes part of the baseline information that provides the foundation of transparency and price discovery in the U.S. securities markets.[197] The Commission therefore believes that it should be included in the definition of core data so that investors and other market participants who consume core data can continue to use it to make informed trading and investment decisions.[198]

To further limit the cost and complexity of the inclusion of odd-lots priced at or better than the NBBO in core data, the definition of core data requires these odd-lots to be represented in the aggregate at each price level at each national securities exchange or national securities association rather than on an order-by-order basis.[199] Finally, as discussed below, the Commission is modifying the proposed definition of round lot, which, relative to the proposal, will reduce the number of round lot tiers and eliminate certain better priced quotation information from the NBBO.[200] However, the inclusion of odd-lot quotes priced at or better than

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197 See Proposing Release, 85 FR at 16736, 16739.

198 See also infra Section II.C.2(c) (discussing why certain other data that is included in SIP data today is not included in core data but will be available through other means).

199 This would not reintroduce a single-price-only odd-lot aggregation methodology in the same sense that prompted concerns from some commenters. See infra note 232 and accompanying text; Section II.C.2(d). Aggregating better-priced odd-lots at each price level at each exchange is not the same as aggregating odd-lots into round lots. Rather, it simply means that better-priced odd-lot orders will be represented in core data in terms of the total number of shares available at each price level at each exchange rather than on an order-by-order basis. For example, if the NBB for XYZ, Inc. is 100 shares at $25.00, and there are three orders of five shares and two orders of ten shares at $25.01 on Exchange A, a competing consolidator's core data product would show 35 shares at $25.01 on Exchange A.

200 See infra Section II.D (stating that increasing the minimum stock price for the first sub-100 share round lot tier from $50 to $250 will not improve odd-lot transparency for stocks priced between $50 and $250). See also supra Tables 1 and 2.